

```
import cv2
import numpy as np
import matplotlib.pyplot as plt
```

```
img = cv2.imread("image.png")

img_rgb = cv2.cvtColor(img, cv2.COLOR_BGR2RGB)

plt.figure(figsize=(6,6))
plt.imshow(img_rgb)
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image



```
gray = cv2.cvtColor(img, cv2.COLOR_BGR2GRAY)

plt.figure(figsize=(6,6))
plt.imshow(gray, cmap='gray')
plt.title("Grayscale Image")
plt.axis("off")
plt.show()
```

## Grayscale Image



```
gaussian = cv2.GaussianBlur(gray,(5,5),0)
```

```
plt.figure(figsize=(6,6))  
plt.imshow(gaussian,cmap='gray')  
plt.title("Gaussian Filter")  
plt.axis("off")  
plt.show()
```

Run cell (Ctrl+Enter)  
cell executed since last change

executed by RITIK SAINI  
10:39 AM (0 minutes ago)  
executed in 0.057s

## Gaussian Filter



```
median = cv2.medianBlur(gray,5)
```

```
plt.figure(figsize=(6,6))  
plt.imshow(median,cmap='gray')  
plt.title("Median Filter")  
plt.axis("off")  
plt.show()
```

## Median Filter



```
average = cv2.blur(gray,(5,5))

plt.figure(figsize=(6,6))
plt.imshow(average,cmap='gray')
plt.title("Average Filter")
plt.axis("off")
plt.show()
```

Run cell (Ctrl+Enter)  
cell executed since last change

executed by RITIK SAINI  
10:39 AM (0 minutes ago)  
executed in 0.057s

## Average Filter



```
laplacian = cv2.Laplacian(gray,cv2.CV_64F)

laplacian = np.uint8(np.absolute(laplacian))

plt.figure(figsize=(6,6))
plt.imshow(laplacian,cmap='gray')
plt.title("Laplacian Edge Detection")
plt.axis("off")
plt.show()
```

## Laplacian Edge Detection



```
sobel_x = cv2.Sobel(gray,  
                    cv2.CV_64F,  
                    1,  
                    0,  
                    ksize=3)  
  
sobel_x = np.uint8(np.absolute(sobel_x))  
  
plt.figure(figsize=(6,6))  
plt.imshow(sobel_x,cmap='gray')  
plt.title("Sobel X")  
plt.axis("off")  
plt.show()
```

Run cell (Ctrl+Enter)  
cell executed since last change  
  
executed by RITIK SAINI  
10:39 AM (0 minutes ago)  
executed in 0.057s

## Sobel X

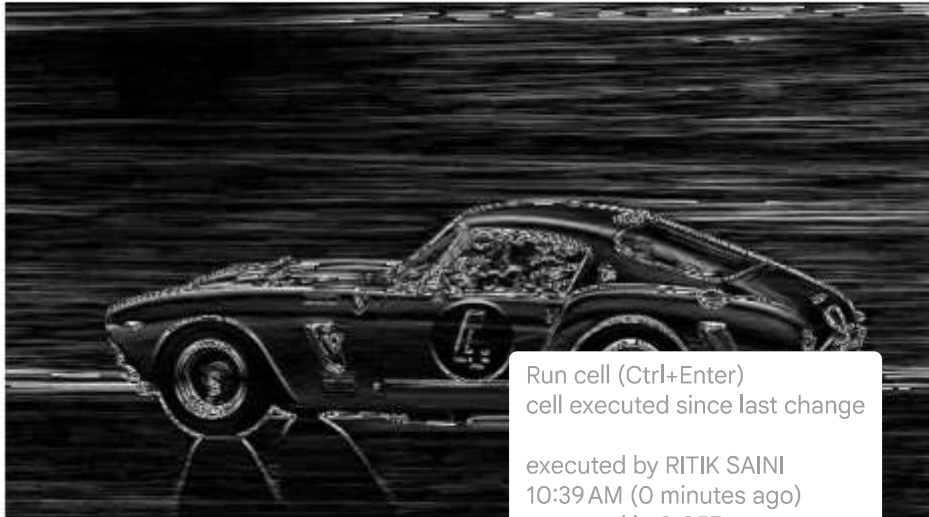


```
sobel_y = cv2.Sobel(gray,  
                    cv2.CV_64F,  
                    0,  
                    1,  
                    ksize=3)
```

```
sobel_y = np.uint8(np.absolute(sobel_y))

plt.figure(figsize=(6,6))
plt.imshow(sobel_y,cmap='gray')
plt.title("Sobel Y")
plt.axis("off")
plt.show()
```

Sobel Y



```
sobel_combined = cv2.bitwise_or(
    sobel_x,
    sobel_y
)

plt.figure(figsize=(6,6))
plt.imshow(sobel_combined,cmap='gray')
plt.title("Combined Sobel")
plt.axis("off")
plt.show()
```

Combined Sobel



```
images = [
```

```

    gray,
    gaussian,
    median,
    average,
    laplacian,
    sobel_combined
]

titles = [
    "Original",
    "Gaussian",
    "Median",
    "Average",
    "Laplacian",
    "Sobel"
]

plt.figure(figsize=(15,10))

for i in range(len(images)):

    plt.subplot(2,3,i+1)

    plt.imshow(images[i],
                cmap='gray')

    plt.title(titles[i])

    plt.axis("off")

plt.tight_layout()
plt.show()

```

Run cell (Ctrl+Enter)  
cell executed since last change

executed by RITIK SAINI  
10:39 AM (0 minutes ago)  
executed in 0.057s



```

print("""
Observations:

```

1. Gaussian Filter reduced noise smoothly.
2. Median Filter removed impulse noise effectively.
3. Average Filter blurred the image more than Gaussian.
4. Laplacian Filter highlighted edges and fine details.

5. Sobel Filter detected horizontal and vertical edges.
  6. Low-pass filters are useful for smoothing.
  7. High-pass filters are useful for edge enhancement.
- ```
"""
```

#### Observations:

1. Gaussian Filter reduced noise smoothly.
2. Median Filter removed impulse noise effectively.
3. Average Filter blurred the image more than Gaussian.
4. Laplacian Filter highlighted edges and fine details.
5. Sobel Filter detected horizontal and vertical edges.
6. Low-pass filters are useful for smoothing.
7. High-pass filters are useful for edge enhancement.

```
print("""
Result:
```

```
Spatial filtering techniques using
Median, Average, Laplacian and Sobel
were successfully implemented using
Low-pass filters reduced noise while
high-pass filters enhanced edges and
important image features.
""")
```

Run cell (Ctrl+Enter)  
cell executed since last change  
executed by RITIK SAINI  
10:39 AM (0 minutes ago)  
executed in 0.057s

#### Result:

Spatial filtering techniques using Gaussian, Median, Average, Laplacian and Sobel filters were successfully implemented using OpenCV.

Low-pass filters reduced noise while high-pass filters enhanced edges and important image features.